

several efforts towards deploying Li-Fi technology.

The Defence Research and Development Organisation (DRDO) and the Defence Innovation Organisation (DIO) are among the organisations facilitating these initiatives. Some of the major programmes are listed in Table 2.

DRDO has developed a secure Li-Fi optical communication system for deployment in submarines, bunkers, missile facilities, and forward bases where RF emission control is required.

The system consists of a transmitter, a confined optical channel, and a receiver. At the transmitting end, data is encrypted and converted into light pulses using a high-intensity LED or laser diode. These pulses are directed into a confined optical channel such as optical fibres, reflective conduits, or sealed line-of-sight corridors to minimise leakage.

The receiver captures the light pulses, demodulates them, and

restores encrypted data streams. These streams are then decrypted and delivered to end users, such as command systems or operational terminals. After laboratory validation, the prototypes are now being evaluated for link stability under varying lighting conditions and for seamless switching between RF and optical channels.

DRDO secure Li-Fi backhaul node. DRDO has developed the secure Li-Fi backhaul node to provide a protected optical link between communication nodes such as command centres, forward bases, and mobile platforms, thereby reducing reliance on RF backhaul channels.

Contested environments

Contested environments are battle spaces where communication systems are under constant threat from adversaries. In such settings, RF networks are liable to be jammed, intercepted, or spoofed, compromising troop positions and disrupting coordination. In dense urban areas, underwater zones, and electronic warfare theatres signal reach is limited, thereby increasing vulnerability. To operate effectively in the contested environments, secure alternatives, like Li-Fi, are required that resist interference and reduce detection risks.

The node confines signals to optical paths, minimising RF signatures in contested environments and enabling covert communication within submarines, bunkers, missile facilities, and hardened shelters. Field trials are underway to assess resilience against ambient light variation, vibration, and RF interference.

Vehicular communication systems (VCS) for defence fleets. Vehicular Li-Fi communication systems for defence fleets have been initiated by the National Institute of Electronics and Information Technology (NIELIT) in partnership with Sri Ramakrishna Institute of Technology.

These VCS allow vehicles to operate under varied conditions and use transmitters embedded in headlights and brake lights to generate encoded optical pulses carrying braking and proximity information. Paired receivers on leading and following vehicles capture these pulses and convert them into digital data streams.

Braking signals synchronise deceleration, while proximity data helps maintain safe distances and fleet formation. The VCS is currently undergoing field testing.

Indoor navigation system for defence forces in GPS-denied environments. DRDO, in partnership

TABLE 1
COMPARISON OF Wi-Fi AND Li-Fi

Feature	Wi-Fi	Li-Fi
Medium of transmission	Radio frequency (RF) waves	Visible light (LED bulbs)
Standard	IEEE 802.11	IEEE 802.15.7
Speed	Up to 1-10Gbps (Wi-Fi 6/7)	Up to 224Gbps (lab conditions)
Range	Works through walls, ~tens of metres	Line-of-sight, ~10 metres
Interference	Susceptible to RF interference	No RF interference; safe in EM-sensitive areas
Security	Easier to intercept	More secure (light cannot penetrate walls)
Availability	Widely deployed globally	Emerging, limited deployment
Applications	Homes, offices, public spaces	Hospitals, aircraft cabins, underwater, and defence
Cost and infrastructure	Mature, low cost	Requires LED infrastructure, higher cost

TABLE 2
SOME MAJOR DEFENCE Li-Fi INITIATIVES IN INDIA

Programme	Sponsoring Body	Support Available for Li-Fi Projects	Projects Availing Support
iDEX (SPARK)	Defence Innovation Organisation (DIO)	Grant for prototype development and challenge-based funding for secure, RF-free systems	Velmenni Li-Fi deployment project for Indian Navy
iDEX Prime	Defence Innovation Organisation (DIO)	Grant for mature, high-impact Li-Fi systems in defence and strategic applications	Velmenni project for providing Li-Fi systems for secure indoor communication in defence environments, particularly naval and tactical use cases
Technology Development Fund (TDF)	Defence Research and Development Organisation (DRDO)	Funding for indigenous R&D in secure optical communication, photonic components, and embedded systems	Recipients not listed publicly for industry-led projects for the development of Li-Fi hardware and secure data transmission modules